Unit circle

1

a $\frac{1}{6}$ **b** $\frac{1}{2}$ **c** $\frac{5}{4}$

2

a Quadrant 4 **b** Quadrant 1 **c** Quadrant 2 **d** Quadrant 3

e Quadrant 1 **f** Quadrant 4 **g** Quadrant 3 **h** Quadrant 2

3 0° or $360^\circ, 90^\circ, 180^\circ, 270^\circ$

4

a

i Quadrant 1 and Quadrant 2 **ii** Quadrant 3 and Quadrant 4

b

i Quadrant 1 and Quadrant 4 **ii** Quadrant 2 and Quadrant 3

c

i Quadrant 1 and Quadrant 3 **ii** Quadrant 2 and Quadrant 4

5

a Quadrant 2 **b** Quadrant 2 **c** Quadrant 2 **d** Quadrant 1

6

a Positive **b** Positive **c** Negative **d** Negative

e Negative **f** Negative **g** Positive **h** Negative

i Negative **j** Negative **k** Positive **l** Positive

m Positive **n** Positive **o** Negative **p** Positive

7

a

i $\frac{4}{5}$ **ii** $\frac{3}{5}$ **iii** $\frac{4}{3}$

b

i $\frac{20}{29}$ **ii** $-\frac{21}{29}$ **iii** $-\frac{20}{21}$

c

i $-\frac{21}{29}$ **ii** $\frac{20}{29}$ **iii** $-\frac{21}{20}$

8

a Yes **b** Yes **c** No **d** Yes

9

a False **b** False **c** True **d** True

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e True

10

a True **b** False **c** True **d** True

e False

11

a $2\sqrt{13}$

b

i $\frac{-3\sqrt{13}}{13}$ **ii** $\frac{-2\sqrt{13}}{13}$ **iii** $\frac{3}{2}$

12

a $\sin\theta = -\frac{4}{5}$ **b** $\tan\theta = -\frac{4}{3}$

13

a $\cos\theta = \pm\frac{3}{4}$ **b** $\tan\theta = \pm\frac{\sqrt{7}}{3}$

14

a $\cos\theta = -\frac{4}{5}$ **b** $\tan\theta = -\frac{3}{4}$

15

a $\cos\theta = \frac{10\sqrt{269}}{269}$ **b** $\sin\theta = -\frac{13\sqrt{269}}{269}$

16 $\cos\theta = -\frac{8}{17}$

17

a $\sin\theta = \frac{11}{61}$ **b** $\sin\theta = \frac{\sqrt{13}}{7}$

18

a $\tan\theta = \frac{2\sqrt{10}}{3}$ **b** $\tan\theta = \frac{1}{3}$

Equivalent ratios

19

a Quadrant 2 **b** Positive **c** 30° **d** $\sin 30^\circ$

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20

a Quadrant 2 b Negative c 30° d $-\cos 30^\circ$

21

a Quadrant 4 b Negative c 30° d $-\tan 30^\circ$

22

a $\sin 87^\circ$ b $-\cos 15^\circ$ c $-\tan 61^\circ$ d $-\sin 60^\circ$

e $\sin 33^\circ$ f $-\tan 50^\circ$ g $\sin 40^\circ$ h $-\cos 65^\circ$

i $-\tan 15^\circ$ j $-\sin 55^\circ$ k $\cos 62^\circ$ l $-\tan 2^\circ$

m $-\sin 41^\circ$ n $-\cos 42^\circ$ o $\tan 71^\circ$ p $\sin 41^\circ$

q $-\cos 77^\circ$ r $-\tan 40^\circ$

23

a $-\frac{b}{a}$ b a c b d $-a$

e $-a$ f $-b$

24

a

i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b

i $\sin 49^\circ$ ii $-\sin 49^\circ$ iii $-\sin 49^\circ$

c

i $\sin x$ ii $-\sin x$ iii $-\sin x$

25

a

i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b

i $-\cos 63^\circ$ ii $-\cos 63^\circ$ iii $\cos 63^\circ$

c

i $-\cos x$ ii $-\cos x$ iii $\cos x$

26

a

i $Q(-a, b)$ ii $R(-a, -b)$ iii $S(a, -b)$

b

i $-\tan 62^\circ$ ii $\tan 62^\circ$ iii $-\tan 62^\circ$

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c

i $-\tan x$ ii $\tan x$ iii $-\tan x$

27

a $\sin\theta$ b $\cos\theta$ c $\sin\theta$ d $-\cos\theta$

e $-\sin\theta$ f $\cos\theta$ g $-\sin\theta$ h $\cos\theta$

28

a 0.93 b 0.36 c -0.93 d 0.36

e -0.36 f -0.93 g 0.36 h -0.93

Use technology

29

a 0.56 b -0.59 c -0.34 d 0.49

e -0.16 f -0.79 g -0.68 h 57.29

i 0.33 j 0.33 k 0.57 l -0.33

30

a 1 b 1 c 1 d 1

31

a $(0.819, 0.574)$ b 73° c $(-0.602, 0.799)$

32

a $(0.41, 0.91)$ b $(-0.41, 0.91)$ c 114° d $(0.41, -0.91)$

e 294°

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